CLASS 10 · ICSE · SAMPLE PAPER SOLUTIONS

MathSetu Sample Paper 1: Mathematics

Step-by-step solutions with the marks for each question

The question numbers match the question paper.

1 Section A, Question 1: choose the correct answer (15 marks)

1. An article is marked at ₹800 and sold within the same state. If the rate of GST is 18%, the CGST charged is: (a) ₹36 (b) ₹72 (c) ₹80 (d) ₹144 1 mark

SOLUTION

For a sale within a state, CGST is half the GST rate: 9% of ₹800 $=$ ₹72.

Answer: (b) ₹72

2. ₹500 is deposited every month for 12 months in a recurring deposit account. The equivalent principal for one month, used to calculate the interest, is: (a) ₹6000 (b) ₹33,000 (c) ₹39,000 (d) ₹78,000 1 mark

SOLUTION

Equivalent principal for one month $= P \times \frac{n(n+1)}{2} = 500 \times \frac{12 \times 13}{2} =$ ₹39,000.

Answer: (c) ₹39,000

3. A ₹100 share is bought at ₹125 and pays a dividend of 10%. The rate of return on the investment is: (a) 8% (b) 10% (c) 12.5% (d) 25% 1 mark

SOLUTION

Dividend on one share $=$ ₹10 on an investment of ₹125, so the return is $\frac{10}{125} \times 100 = 8\%$.

Answer: (a) 8%

4. The solution set of $-3 < 2x - 1 \leq 5$, where x is a whole number, is: (a) $\{1, 2, 3\}$ (b) $\{0, 1, 2\}$ (c) $\{0, 1, 2, 3\}$ (d) $\{-1, 0, 1, 2, 3\}$ 1 mark

SOLUTION

Add 1 throughout: $-2 < 2x \leq 6$, so $-1 < x \leq 3$. The whole numbers in this range are 0, 1, 2, 3.

Answer: (c) $\{0, 1, 2, 3\}$

5. The roots of $x^2 - 6x + 9 = 0$ are: (a) real and unequal (b) real and equal (c) not real (d) equal in size and opposite in sign **1 mark**

SOLUTION

$D = (-6)^2 - 4(1)(9) = 0$, so the roots are real and equal (both are 3).

Answer: (b) real and equal

6. The mean proportional between 4 and 25 is: (a) 10 (b) 14.5 (c) 21 (d) 100 **1 mark**

SOLUTION

Mean proportional $= \sqrt{4 \times 25} = 10$.

Answer: (a) 10

7. The remainder when $x^3 - 2x^2 + 3x - 4$ is divided by $x - 2$ is: (a) -4 (b) 0 (c) 2 (d) 6 **1 mark**

SOLUTION

By the Remainder Theorem, the remainder is $f(2) = 8 - 8 + 6 - 4 = 2$.

Answer: (c) 2

8. A is a matrix of order 2×3 and B is a matrix of order 3×1. The order of AB is: (a) 3×3 (b) 2×1 (c) 1×2 (d) AB is not possible **1 mark**

SOLUTION

The columns of A (3) equal the rows of B (3), so AB exists, with the rows of A and the columns of B: 2×1.

Answer: (b) 2×1

9. The nth term of an A.P. is $5n - 2$. Its common difference is: (a) -2 (b) 2 (c) 3 (d) 5 **1 mark**

SOLUTION

$T_1 = 3$ and $T_2 = 8$, so $d = 8 - 3 = 5$.

Answer: (d) 5

10. The 5th term of the G.P. $3, 6, 12, \ldots$ is: (a) 24 (b) 36 (c) 48 (d) 96 **1 mark**

SOLUTION

$a = 3, r = 2$: $T_5 = ar^4 = 3 \times 16 = 48$.

Answer: (c) 48

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11. The image of the point $(3, -4)$ after reflection in the y-axis is: (a) $(3, 4)$ (b) $(-3, 4)$ (c) $(-3, -4)$ (d) $(-4, 3)$ 1 mark

SOLUTION

Reflection in the y-axis changes the sign of the x-coordinate only: $(3, -4) \to (-3, -4)$.

Answer: (c) $(-3, -4)$

12. The slope of the line $3x - 4y + 7 = 0$ is: (a) $\frac{3}{4}$ (b) $-\frac{3}{4}$ (c) $\frac{4}{3}$ (d) $\frac{7}{4}$ 1 mark

SOLUTION

Write it as $y = \frac{3}{4}x + \frac{7}{4}$. The slope is $\frac{3}{4}$.

Answer: (a) $\frac{3}{4}$

13. An arc of a circle subtends an angle of 110° at the centre. The angle it subtends at a point on the remaining part of the circle is: (a) 55° (b) 70° (c) 110° (d) 220° 1 mark

SOLUTION

The angle at the centre is twice the angle at the circumference: $\frac{110^\circ}{2} = 55^\circ$.

Answer: (a) 55°

14. The volume of a cone of radius 3 cm and height 7 cm is: (a) 22 cm^3 (b) 66 cm^3 (c) 132 cm^3 (d) 198 cm^3 1 mark

SOLUTION

Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 9 \times 7 = 66$ cm^3.

Answer: (b) 66 cm^3

15. A letter is chosen at random from the letters of the word TRIANGLE. The probability that it is a vowel is: (a) $\frac{1}{4}$ (b) $\frac{3}{8}$ (c) $\frac{1}{2}$ (d) $\frac{5}{8}$ 1 mark

SOLUTION

The word has 8 letters, of which 3 (I, A, E) are vowels. Probability $= \frac{3}{8}$.

Answer: (b) $\frac{3}{8}$

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2 Section A, Question 2 (12 marks)

16. Meera deposits ₹1200 every month for 2 years in a recurring deposit account that pays simple interest at 8% per annum. Find the interest she earns and the amount she gets on maturity. 4 marks

SOLUTION

$P = ₹1200$, $n = 24$ months, $r = 8\%$.

Interest $= P \times \frac{n(n+1)}{2 \times 12} \times \frac{r}{100} = 1200 \times \frac{24 \times 25}{24} \times \frac{8}{100} = ₹2400.$

Total deposited $= 1200 \times 24 = ₹28{,}800$, so the maturity amount $= 28{,}800 + 2400 = ₹31{,}200.$

Answer: interest ₹2400, maturity amount ₹31,200

17. If $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$, find $A^2 - 2B$. 4 marks

SOLUTION

$$A^2 = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 4-1 & 2+3 \\ -2-3 & -1+9 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix}$$

$$2B = \begin{bmatrix} 2 & 0 \\ 4 & -2 \end{bmatrix}$$

$$A^2 - 2B = \begin{bmatrix} 3-2 & 5-0 \\ -5-4 & 8+2 \end{bmatrix}$$

Answer: $\begin{bmatrix} 1 & 5 \\ -9 & 10 \end{bmatrix}$

18. Prove that $\frac{\cos A}{1 - \sin A} + \frac{\cos A}{1 + \sin A} = 2 \sec A.$ 4 marks

SOLUTION

$$\text{LHS} = \frac{\cos A\,(1 + \sin A) + \cos A\,(1 - \sin A)}{(1 - \sin A)(1 + \sin A)}$$

$$= \frac{2 \cos A}{1 - \sin^2 A} = \frac{2 \cos A}{\cos^2 A}$$

$$= \frac{2}{\cos A} = 2 \sec A = \textbf{RHS. Proved.}$$

3 Section A, Question 3 (13 marks)

19. A solid metal sphere of radius 6 cm is melted and recast into a solid cylinder of radius 4 cm. Find the height of the cylinder and its curved surface area, correct to one decimal place. 4 marks

SOLUTION

Volume of the sphere $= \frac{4}{3}\pi \times 6^3 = 288\pi$ cm^3.

Volume of the cylinder $= \pi \times 4^2 \times h = 16\pi h$. So $16\pi h = 288\pi$ and $h = 18$ cm.

Curved surface area $= 2\pi r h = 2 \times \frac{22}{7} \times 4 \times 18 = \frac{3168}{7} \approx 452.6$ cm^2.

Answer: height 18 cm, curved surface area ≈ 452.6 cm^2

20. $A(2, 3)$ and $B(6, -5)$ are two points. Find (i) the slope of AB (ii) the equation of the perpendicular bisector of AB (iii) the point where this bisector meets the x-axis. 4 marks

SOLUTION

(i) Slope of $AB = \dfrac{-5-3}{6-2} = -2$.

(ii) The mid-point of AB is $\left(\dfrac{2+6}{2}, \dfrac{3-5}{2}\right) = (4, -1)$, and the slope of the perpendicular bisector is $\dfrac{1}{2}$.

Its equation is $y + 1 = \dfrac{1}{2}(x - 4)$, that is $x - 2y - 6 = 0$.

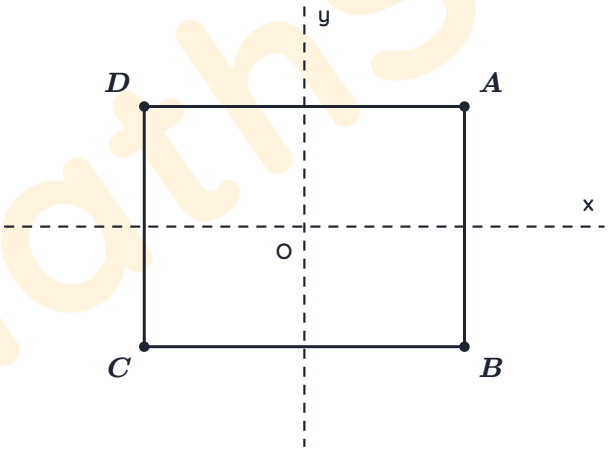
(iii) Put $y = 0$: $x = 6$. The point is $(6, 0)$.

Answer: (i) -2 **(ii)** $x - 2y - 6 = 0$ **(iii)** $(6, 0)$

21. Use graph paper for this question, with 1 cm $=$ 1 unit on both axes. Plot the point $A(4, 3)$. Reflect A in the x-axis to get B, reflect A in the origin to get C, and reflect A in the y-axis to get D. (i) Write the coordinates of B, C and D. (ii) Name the figure $ABCD$. (iii) Find its perimeter and its area. 5 marks

SOLUTION

(i) $B = (4, -3)$, $C = (-4, -3)$ and $D = (-4, 3)$.



(ii) AB and CD are vertical, BC and DA are horizontal, and all the angles are right angles, so $ABCD$ is a rectangle.

(iii) $AB = 6$ units and $BC = 8$ units. Perimeter $= 2(6 + 8) = 28$ units and area $= 6 \times 8 = 48$ square units.

Answer: (i) $B(4, -3)$, $C(-4, -3)$, $D(-4, 3)$ **(ii) a rectangle (iii) perimeter 28 units, area 48 square units**

4 Section B, Question 4 (10 marks)

22. A dealer in Delhi sells a washing machine, marked at ₹20,000, to a customer in Delhi at a discount of 10%. The rate of GST is 18%. Find the CGST, the SGST and the total amount the customer pays. 3 marks

SOLUTION

Selling price after discount $= 20{,}000 - 10\%$ of $20{,}000 =$ ₹18,000.

CGST $= 9\%$ of 18,000 $=$ ₹1620 and SGST $= 9\%$ of 18,000 $=$ ₹1620.

Amount paid $= 18{,}000 + 1620 + 1620 =$ ₹21,240.

Answer: CGST ₹1620, SGST ₹1620, amount paid ₹21,240

23. Solve $x^2 - 5x - 3 = 0$, giving the answers correct to two decimal places. 3 marks

SOLUTION

$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \dfrac{5 \pm \sqrt{25 + 12}}{2} = \dfrac{5 \pm \sqrt{37}}{2}$, and $\sqrt{37} \approx 6.083$.

$x = \dfrac{5 + 6.083}{2} \approx 5.54$ or $x = \dfrac{5 - 6.083}{2} \approx -0.54$.

Answer: $x = 5.54$ **or** $x = -0.54$

24. Find the mean of the following distribution by the step-deviation method. 4 marks

Class	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
Frequency	6	10	16	12	6

SOLUTION

Take the assumed mean $A = 50$ and the class width $i = 20$, with $t = \dfrac{x - A}{i}$.

Class	f	x	t	ft
0 – 20	6	10	-2	-12
20 – 40	10	30	-1	-10
40 – 60	16	50	0	0
60 – 80	12	70	1	12
80 – 100	6	90	2	12
Total	50			2

Mean $= A + \dfrac{\sum ft}{\sum f} \times i = 50 + \dfrac{2}{50} \times 20 = 50.8$.

Answer: mean $= 50.8$

5 Section B, Question 5 (10 marks)

25. A man invests ₹9600 in ₹100 shares at a premium of 20%. The company declares a dividend of 15%. Find (i) the number of shares he buys (ii) his annual dividend (iii) the rate of return on his investment. 3 marks

SOLUTION

(i) Market value of one share $= 100 + 20 =$ ₹120. Number of shares $= \dfrac{9600}{120} = 80$.

(ii) Dividend on one share $= 15\%$ of ₹100 $=$ ₹15. Annual dividend $= 80 \times 15 =$ ₹1200.

(iii) Rate of return $= \dfrac{1200}{9600} \times 100 = 12.5\%$.

Answer: (i) 80 shares (ii) ₹1200 (iii) 12.5%

26. In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 4)^\circ$, $\angle B = (x + 40)^\circ$ and $\angle C = (3x + 16)^\circ$. Find x and all four angles. 3 marks

SOLUTION

Opposite angles of a cyclic quadrilateral add up to 180°: $(2x + 4) + (3x + 16) = 180$, so $5x = 160$ and $x = 32$.

$\angle A = 68^\circ$, $\angle C = 112^\circ$, $\angle B = 72^\circ$ and $\angle D = 180^\circ - 72^\circ = 108^\circ$.

Answer: $x = 32$; the angles are 68°, 72°, 112° and 108°

27. Use the Factor Theorem to show that $x - 2$ is a factor of $2x^3 + x^2 - 13x + 6$. Hence factorise the expression completely. 4 marks

SOLUTION

Let $f(x) = 2x^3 + x^2 - 13x + 6$. Then $f(2) = 16 + 4 - 26 + 6 = 0$, so $x - 2$ is a factor.

Dividing $f(x)$ by $x - 2$ gives the quotient $2x^2 + 5x - 3$.

$2x^2 + 5x - 3 = 2x^2 + 6x - x - 3 = 2x(x + 3) - 1(x + 3) = (2x - 1)(x + 3)$.

Answer: $(x - 2)(2x - 1)(x + 3)$

6 Section B, Question 6 (10 marks)

28. Solve $3x - 7 \leq 7x + 5 < 19$, where x is an integer, and show the solution set on a number line. 3 marks

SOLUTION

From $3x - 7 \leq 7x + 5$: $-12 \leq 4x$, so $x \geq -3$. From $7x + 5 < 19$: $7x < 14$, so $x < 2$.

So $-3 \leq x < 2$, and the solution set is $\{-3, -2, -1, 0, 1\}$. On the number line these five integers are marked with dots.



Answer: $\{-3, -2, -1, 0, 1\}$

29. The 4th term of an A.P. is 22 and its 15th term is 66. Find the first term, the common difference and the sum of its first 8 terms. 3 marks

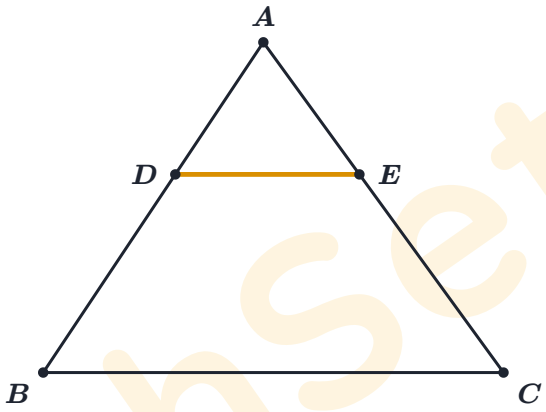
SOLUTION

$a + 3d = 22$ and $a + 14d = 66$. Subtracting, $11d = 44$, so $d = 4$ and $a = 10$.

$S_8 = \frac{8}{2}[2 \times 10 + 7 \times 4] = 4 \times 48 = 192$.

Answer: $a = 10$, $d = 4$, $S_8 = 192$

30. In $\triangle ABC$, D lies on AB and E lies on AC so that $DE \parallel BC$ and $AD : DB = 2 : 3$. (i) Prove that $\triangle ADE \sim \triangle ABC$. (ii) If $DE = 4$ cm, find BC. (iii) Find the ratio of the area of $\triangle ADE$ to the area of trapezium $DBCE$. 4 marks



SOLUTION

(i) $\angle A$ is common and $\angle ADE = \angle ABC$ (corresponding angles, $DE \parallel BC$). So $\triangle ADE \sim \triangle ABC$ (AA).

(ii) $\frac{AD}{AB} = \frac{2}{2+3} = \frac{2}{5}$, so $\frac{DE}{BC} = \frac{2}{5}$ and $BC = \frac{5}{2} \times 4 = 10$ cm.

(iii) $\frac{\text{area of } \triangle ADE}{\text{area of } \triangle ABC} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$. So the trapezium has $25 - 4 = 21$ of the 25 parts.

Answer: (ii) $BC = 10$ **cm (iii)** $4 : 21$

7 Section B, Question 7 (10 marks)

31. What number must be added to each of 2, 6, 8 and 18 so that the four results are in proportion? 3 marks

SOLUTION

Let the number be x. Then $\frac{2+x}{6+x} = \frac{8+x}{18+x}$.

$(2+x)(18+x) = (6+x)(8+x)$ gives $36 + 20x + x^2 = 48 + 14x + x^2$, so $6x = 12$ and $x = 2$.

Check: 4, 8, 10, 20 are in proportion, since $4 \times 20 = 8 \times 10$.

Answer: 2

32. The 3rd term of a G.P. is 12 and its 6th term is 96. Find the first term, the common ratio and the sum of its first 8 terms. 3 marks

SOLUTION

$ar^2 = 12$ and $ar^5 = 96$. Dividing, $r^3 = 8$, so $r = 2$ and $a = 3$.

$$S_8 = \frac{a(r^8 - 1)}{r - 1} = \frac{3(256 - 1)}{2 - 1} = 765.$$

Answer: $a = 3$, $r = 2$, $S_8 = 765$

33. From a point A on level ground, the angle of elevation of the top of a tower is 30°. On walking 40 m towards the tower to a point B, the angle of elevation becomes 60°. Find the height of the tower, correct to two decimal places, and the distance of B from the tower. (Take $\sqrt{3} = 1.732$.) 4 marks

SOLUTION

Let the tower be h m high and let B be x m from its foot.

From B: $\tan 60^\circ = \frac{h}{x}$, so $h = \sqrt{3}\,x$. From A: $\tan 30^\circ = \frac{h}{x + 40}$, so $x + 40 = \sqrt{3}\,h$.

Then $x + 40 = \sqrt{3} \times \sqrt{3}\,x = 3x$, so $x = 20$ m.

$h = 20\sqrt{3} = 20 \times 1.732 = 34.64$ m.

Answer: height 34.64 m; B is 20 m from the tower

8 Section B, Question 8 (10 marks)

34. Cards numbered 1 to 30 are put in a box and one card is drawn at random. Find the probability that the number on it is (i) a multiple of 4 (ii) a prime number (iii) a perfect square. 3 marks

SOLUTION

(i) Multiples of 4: 4, 8, 12, 16, 20, 24, 28, which is 7 cards. Probability $= \frac{7}{30}$.

(ii) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, which is 10 cards. Probability $= \frac{10}{30} = \frac{1}{3}$.

(iii) Perfect squares: 1, 4, 9, 16, 25, which is 5 cards. Probability $= \frac{5}{30} = \frac{1}{6}$.

Answer: (i) $\frac{7}{30}$ (ii) $\frac{1}{3}$ (iii) $\frac{1}{6}$

35. In what ratio does the y-axis divide the line segment joining $A(-4, 1)$ and $B(6, 6)$? Also find the point of division. 3 marks

SOLUTION

Let the ratio be $k : 1$. The x-coordinate of the point of division is 0: $\frac{6k - 4}{k + 1} = 0$, so $k = \frac{2}{3}$. The ratio is $2 : 3$.

$$y = \frac{2 \times 6 + 3 \times 1}{2 + 3} = \frac{15}{5} = 3.$$

Answer: ratio $2 : 3$, point $(0, 3)$

36. A conical tent is 12 m high and the radius of its base is 9 m. Find (i) the cost of the canvas for its curved surface at ₹70 per m^2 (ii) the volume of air inside it, correct to the nearest m^3. 4 marks

SOLUTION

Slant height $l = \sqrt{9^2 + 12^2} = 15$ m.

(i) Curved surface area $= \pi r l = \frac{22}{7} \times 9 \times 15 = \frac{2970}{7}$ m^2. Cost $= \frac{2970}{7} \times 70 =$ ₹29,700.

(ii) Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 81 \times 12 = \frac{7128}{7} \approx 1018$ m^3.

Answer: (i) ₹29,700 (ii) about 1018 m^3

9 Section B, Question 9 (10 marks)

37. The length of a rectangular garden is 4 m more than its breadth, and its area is 96 m^2. Find its length, its breadth and its perimeter. 3 marks

SOLUTION

Let the breadth be x m. Then $x(x+4) = 96$, so $x^2 + 4x - 96 = 0$.

$(x+12)(x-8) = 0$, so $x = 8$ (a length cannot be negative).

Breadth $= 8$ m, length $= 12$ m, perimeter $= 2(8+12) = 40$ m.

Answer: length 12 m, breadth 8 m, perimeter 40 m

38. (i) Chords AB and CD of a circle meet at a point P inside the circle. If $AP = 6$ cm, $PB = 4$ cm and $CP = 3$ cm, find CD. (ii) From a point T outside another circle, a tangent TQ and a secant TRS are drawn, with $TR = 4$ cm and $RS = 5$ cm. Find TQ. 3 marks

SOLUTION

(i) $AP \times PB = CP \times PD$, so $6 \times 4 = 3 \times PD$ and $PD = 8$ cm. Then $CD = 3 + 8 = 11$ cm.

(ii) $TQ^2 = TR \times TS = 4 \times (4+5) = 36$, so $TQ = 6$ cm.

Answer: (i) $CD = 11$ cm (ii) $TQ = 6$ cm

39. The marks of 11 students in a test are 12, 18, 25, 9, 30, 21, 15, 27, 33, 6 and 24. Find (i) the median (ii) the lower quartile (iii) the upper quartile (iv) the interquartile range. 4 marks

SOLUTION

In ascending order: 6, 9, 12, 15, 18, 21, 24, 25, 27, 30, 33. Here $n = 11$.

(i) Median $= \left(\frac{n+1}{2}\right)$th term $=$ 6th term $= 21$.

(ii) Lower quartile $Q_1 = \left(\frac{n+1}{4}\right)$th term $=$ 3rd term $= 12$.

(iii) Upper quartile $Q_3 = \left(\frac{3(n+1)}{4}\right)$th term $=$ 9th term $= 27$.

(iv) Interquartile range $= Q_3 - Q_1 = 27 - 12 = 15$.

Answer: (i) 21 (ii) 12 (iii) 27 (iv) 15

10 Section B, Question 10 (10 marks)

40. Find the equation of the line that passes through $(2, -1)$ and is parallel to the line $3x + 4y = 7$. Also write its y-intercept. 3 marks

SOLUTION

The slope of $3x + 4y = 7$ is $-\frac{3}{4}$, and a parallel line has the same slope.

$y + 1 = -\frac{3}{4}(x - 2)$ gives $4y + 4 = -3x + 6$, that is $3x + 4y - 2 = 0$.

Put $x = 0$: $y = \frac{1}{2}$.

Answer: $3x + 4y - 2 = 0$**, with** y**-intercept** $\frac{1}{2}$

41. Find x and y if $\begin{bmatrix} 2 & x \\ y & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$. 3 marks

SOLUTION

Multiplying, $\begin{bmatrix} 2 + 2x \\ y + 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$.

So $2 + 2x = 8$ gives $x = 3$, and $y + 6 = 7$ gives $y = 1$.

Answer: $x = 3, y = 1$

42. Using a ruler and compasses only, construct $\triangle ABC$ with $AB = 5$ cm, $BC = 6$ cm and $\angle ABC = 60°$. Construct the incircle of the triangle and measure its radius. 4 marks

SOLUTION

Draw $BC = 6$ cm. At B construct an angle of $60°$ and cut off $BA = 5$ cm on its arm. Join AC.

Construct the bisectors of $\angle B$ and $\angle C$. They meet at the incentre I.

From I draw IN perpendicular to BC, meeting it at N. With centre I and radius IN draw the circle. It touches all three sides.

On measuring, IN comes to about 1.6 cm.

Answer: the radius of the incircle measures about 1.6 cm

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